TYLER P. QUIGLEY, PH.D.

Life Sciences | Scientific Evaluation & Diligence | Science Writing | AI for Biology
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Life sciences Ph.D. who evaluates and communicates science across research, intellectual property, investment, and AI. Experience spans scientific due diligence and IP analysis, running evaluation and dealflow for a decentralized research organization, assessing AI systems that perform scientific work, and translating complex research for expert and public audiences. Enters unfamiliar fields quickly and turns ambiguous evidence into clear, decision-useful output.

EDUCATION

Ph.D., Animal Behavior

Arizona State University | April 2024

Research focus in cellular neurobiology: structure and permeability of the honeybee blood-brain barrier across age (electron microscopy, NanoSIMS, stereology).

B.S., Biology

Roanoke College | 2015

PROFESSIONAL EXPERIENCE

Scientific Evaluation Contractor

Edison Scientific and Office Hours | 2025 – Present

- Evaluate outputs from AI systems performing scientific work, identifying errors in reasoning, method, and interpretation across biology and chemistry.
- Generate and review evaluation and training tasks, and provide expert baselining used to measure and improve model performance.

Scientific Advisor

Stealth Early-Stage Biotech (Women's Health) | 2025 – Present

- Advise an early-stage biotech on preclinical strategy; contributed to lead-candidate selection, IND-enabling study design, and an academic collaboration.

Founder

Bayside AI | 2025 – Present

- Build AI tools and automations for small businesses, from scoping through delivery.

Director, Science & Dealflow

PsyDAO | Jan 2025 – Apr 2026

- Built and ran a decentralized funding system for open-science and public-goods research in mental health therapeutics, spanning sourcing, scientific diligence, distributed review, and funding decisions.
- Owned the scientific evidence bar across 20+ early-stage proposals; designed evaluation frameworks, review processes, and decision criteria; advanced three programs to funding.
- Designed and operated a research fellowship program end to end; coordinated cross-functional review across scientific, legal, and community stakeholders and reported to leadership.

Consulting Scientist

Beeard.ai | Jan 2025 – Dec 2025

- Evaluated outputs from an agentic biology platform, identifying failure modes in reasoning across neuroscience, molecular biology, and chemistry, and fed structured findings back to improve the product.

IP Technical Specialist, Life Sciences

Calyx Law | Apr 2024 – Dec 2024

- Evaluated biotech inventions for technical feasibility, mechanistic plausibility, and competitive positioning across mental health therapeutics, longevity, AI, and emerging biotechnologies.
- Conducted prior art and freedom-to-operate analyses and drafted patent and technical documentation for legal and business audiences.

IP Research Assistant

Calyx Law | Mar 2021 – Apr 2023, Sept 2023 – Apr 2024

- Supported 25+ patent applications across chemistry, treatment methods, food science, and software through prior-art and literature research; ran five freedom-to-operate searches and supported multiple prosecution cases.

Technology Transfer Fellow

Skysong Innovations, Arizona State University | Apr – Aug 2023

- Evaluated life sciences inventions for technical merit, commercial pathways, and IP potential; supported invention disclosure intake and triage.

Doctoral Researcher, Animal Behavior & Neuroscience

Arizona State University | Aug 2015 – Apr 2024

- Characterized the honeybee blood-brain barrier end to end, adapting NanoSIMS from geochemistry to biology and building imaging and analysis workflows where no protocol existed.
- Authored peer-reviewed publications and SOPs; secured $20K+ in competitive grants; mentored and trained six researchers.

SELECTED PROJECTS

OpenClaw (Agentic AI Orchestration)

- Built a model-agnostic orchestration harness that combines multiple LLMs for literature synthesis, landscape mapping, and first-pass evaluation.

Honeybee BBB Image-Analysis Pipeline

- Built an image-analysis pipeline automating segmentation and analysis of electron-microscopy datasets from doctoral research.

SELECTED PUBLICATIONS

Quigley, T.P., & Amdam, G.V. (2026). Effect of aging and Varroa parasitism on the paracellular and transcellular permeability of the honeybee blood-brain barrier. PLOS ONE, 21(3), e0343142.

Quigley, T.P., & Amdam, G.V. (2024). Ultrastructural organization of the honeybee blood-brain barrier and comparison with age. bioRxiv preprint. DOI: 10.1101/2024.09.27.615080

Quigley, T.P., & Amdam, G.V. Tracking vitellogenin across the honeybee blood-brain barrier with nanoscale secondary ion mass spectrometry. Manuscript in preparation.

Quigley, T.P., & Amdam, G.V. (2021). Social modulation of ageing: mechanisms, ecology, evolution. Philosophical Transactions of the Royal Society B, 376(1823), 20190738.

Quigley, T.P., Amdam, G.V., & Harwood, G. (2019). Honey bees as bioindicators of changing global agricultural landscapes. Current Opinion in Insect Science, 35, 132-137.

SCIENCE WRITING & COMMUNICATION

The Tab Psychedelic Science Newsletter

psychedelicscience.substack.com | Mar 2021 – Present

- Own the full editorial process for an independent publication; 33 articles, 1,000+ subscribers, 33,000+ all-time views, and open rates consistently above 45%.

Research & Writing Freelancer

Beckley Retreats | 2022

- Science writing on the bioactivity of pharmaceutical compounds, research methodologies, and emerging therapeutic applications.

Science Writer & Public Engagement

Ask A Biologist, Arizona State University | 2016 – 2018

- Managed the Dr. Biology reader Q&A pipeline (assigning, answering, and editing contributor responses) and developed a science-education curriculum for general and K-12 audiences.

SKILLS

Scientific Evaluation & Diligence: Structured scientific review, experimental-design and statistical-validity assessment, mechanistic plausibility, evidence synthesis, competitive landscape analysis, failure-mode identification in AI outputs

Investment & IP: Opportunity and dealflow evaluation, risk-adjusted and probability-of-approval reasoning, prior art and freedom-to-operate analysis, patent drafting, claim substantiation

Operations & Program Management: Process and framework design, dealflow and pipeline tracking, milestone tracking and reporting, cross-functional coordination, program and fellowship design

Writing & Communication: Peer-reviewed publication, science journalism and newsletters, editorial pipeline management, cross-disciplinary and plain-language translation

AI & Computational: Agentic workflow orchestration (OpenClaw), Claude Code, Anthropic API, evaluation of AI-generated scientific reasoning, R, image-analysis pipelines

Laboratory & Instrumentation: Electron microscopy (TEM, SEM, array tomography), NanoSIMS, stereology, sample preparation, assay development, SOP authoring

Domain: Cellular neurobiology, blood-brain barrier biology, aging and longevity biology, mental health therapeutics and CNS pharmacology